PROJECT REPORT

Fake News Detection System using Prolog (Rule-Based AI)

1. Abstract

The rapid growth of digital media and online communication platforms has significantly increased the spread of fake news. This project presents a Fake News Detection System developed using a rule-based approach in Prolog. Unlike machine learning-based systems, this model relies on logical reasoning and predefined rules to classify news as Real, Fake, or Uncertain. The project demonstrates the application of symbolic artificial intelligence techniques in solving a real-world problem.

2. Introduction

Fake news refers to misleading or false information presented as legitimate news. With the widespread use of social media platforms such as WhatsApp, Instagram, and online blogs, misinformation spreads quickly and can have serious consequences.

Manual verification of news is not practical due to the volume of information generated daily. Therefore, an automated system is required to assist users in identifying fake news. This project uses Prolog, a logic programming language, to implement a rule-based system that classifies news based on textual patterns and source credibility.

3. Problem Statement

The problem addressed in this project is:

How can a system automatically determine whether a piece of news is fake or real using logical reasoning instead of statistical machine learning methods?

4. Objectives

- To design a system that classifies news as Fake, Real, or Uncertain
- To implement a rule-based artificial intelligence system using Prolog
- To demonstrate knowledge representation using facts and rules
- To simulate decision-making using logical inference

5. Significance of the Problem

Fake news has become a major concern in the digital age. Its impact includes:

- Spreading misinformation among the public
- Creating panic and confusion
- Influencing opinions and decisions
- Enabling fraud and scams

An effective fake news detection system can help reduce these risks and promote responsible information consumption.

6. Methodology

This project uses a symbolic AI approach based on rules and logical inference rather than machine learning.

6.1 Knowledge Base

The system stores:

- A list of suspicious keywords commonly found in fake news
- A list of trusted news sources

6.2 Rules

The following rules are applied:

- If the news text contains suspicious keywords and the source is not trusted, it is classified as Fake News
- If the source is trusted and no suspicious keywords are present, it is classified as Real News
- In all other cases, the result is classified as Uncertain

6.3 Inference Mechanism

- Pattern matching is performed using substring checks
- Logical negation is used to handle conditions
- Decision-making is based on rule evaluation

7. Implementation

Tools Used

- SWI-Prolog
- Text editor (such as VS Code or Notepad)

System Description

The system takes two inputs:

- News text
- Source of the news

Based on predefined rules, it returns one of the following outputs:

- Fake News
- Real News
- Uncertain

8. Test Cases

News Text	Source	Output
shocking viral news click here	random_blog	Fake News
government releases new policy	bbc	Real News
new technology launched	unknown	Uncertain

9. Key Decisions Made

- Selected Prolog to demonstrate symbolic AI instead of machine learning
- Used rule-based logic for transparency and simplicity
- Chose keywords and source credibility as classification factors
- Introduced an Uncertain category for ambiguous cases

10. Challenges Faced

- Limited natural language processing capabilities in Prolog
- Difficulty in covering all possible fake news patterns
- Lack of built-in machine learning libraries
- Balancing between simplicity and effectiveness

11. Learning Outcomes

This project helped in understanding:

- Logic programming and Prolog syntax
- Differences between symbolic AI and machine learning
- Knowledge representation using facts and rules
- Real-world problem solving using AI techniques
- Designing and testing an intelligent system

12. Future Scope

- Integration with machine learning models for improved accuracy
- Use of advanced natural language processing techniques
- Development of a graphical user interface or web application
- Real-time analysis of news data
- Expansion of the knowledge base

13. Conclusion

The Fake News Detection System developed in this project successfully demonstrates the use of rule-based artificial intelligence using Prolog. Although it does not use statistical learning methods, it effectively applies logical reasoning to classify news. The project highlights the importance of symbolic AI in solving real-world problems and provides a foundation for further enhancements.

14. References

- SWI-Prolog Documentation
- Artificial Intelligence course materials
- Online resources on fake news and misinformation